

OU fits to the gain-modulated RNNs: RT-and-choice fits are leaky at every gain; the same accumulator fitted to choice given the evidence recovers the transition

Ivan Grahek, 24 September 2026. Summary of round 2 plus the threshold sweep; full write-up docs/OU_fits_to_gain_RNNs_round2.pdf, code and RESULTS.md on branch round2-report of rnn_hssm. Prediction: gain 0.8 / 1.0 / 1.2 = leaky / near-perfect / unstable integration, so the OU leak should go positive / zero / negative. Convention throughout: ssm-simulators' update is $x += (v - g \cdot x) \cdot dt + \text{noise}$, so **positive g is leaky**, $g = -\lambda$ in Brunton's notation.

1. Control: HSSM OU fitted to RT and choice (20 networks, bound 1.5 on dv)

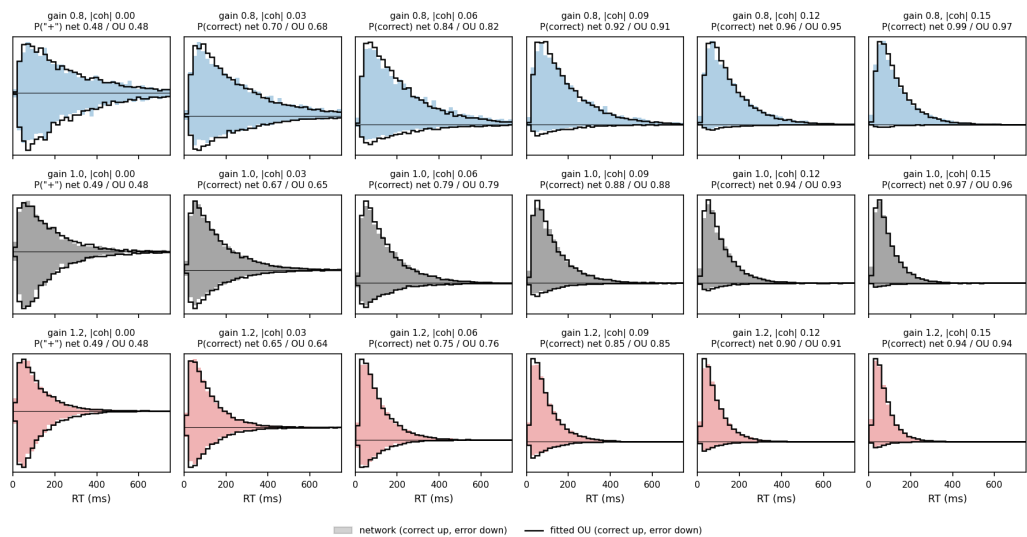


Figure 1. The fit reproduces RT and choice. Network (filled) vs fitted OU (line), by [coherence] and gain, correct up and error down, areas summing to one per panel. Setup: RTs $\times 10$, non-decision time fixed at offset + min RT, no lapse, $v \sim 1 + \text{signed coherence}$ (identity link), a, z, g shared across coherences; 4 chains, $R\text{-hat} \leq 1.01$, one divergence in 600 000 draws. Median RT-quantile error 6.5 / 1.6 / 0.8 ms at gains 0.8 / 1.0 / 1.2.

gain	g native (per s) [94 % HDI]	posterior mass at the +1 ceiling	networks with g > 0 (n = 20)	kernel slope: network vs fitted OU with bound / without
0.8	+10.0 [+9.9, +10.0]	1.00	20 of 20	+0.039 vs -0.041 / +0.068
1.0	+10.0 [+9.9, +10.0]	0.99	20 of 20	+0.002 vs -0.057 / +0.067
1.2	+9.2 [+8.7, +9.7]	0.01	20 of 20	-0.026 vs -0.064 / +0.066

Table 1. Leaky at every gain. g sits on the likelihood's +1 edge at gains 0.8 and 1.0 for every stretch $k = 8 \dots 24$ (native value = k), so only the sign is identified. Recovery: with the theoretical g planted in simulated data the same pipeline returns $g < 0$ at gains 1.0 and 1.2 (sign 6 of 6). Last column: Figure 2.

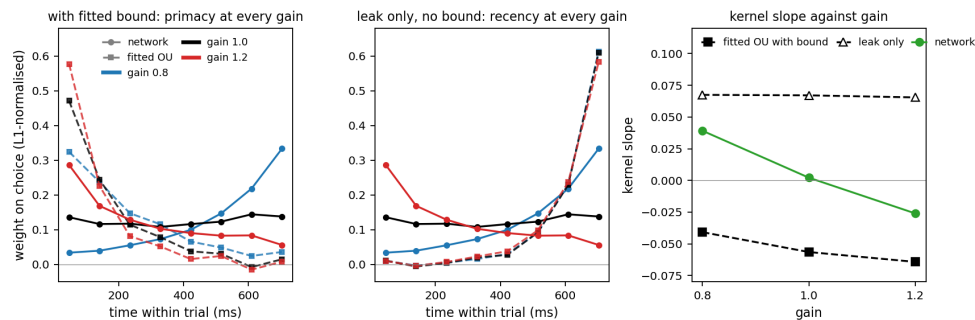


Figure 2. The same fitted model driven by the networks' own evidence streams gets the psychophysical kernel wrong at every gain. With its fitted bound (0.3, reached within 100 ms on > 99.9 % of trials) it commits early: primacy everywhere. With the bound removed the fitted leak (+10 per s) forgets: recency everywhere. The networks go recency $\rightarrow$ flat $\rightarrow$ primacy.

2. Raising the threshold: where the slow dynamics govern the crossing

Every RT here is a threshold we place on dv (the networks read out at 750 ms). At 1.5 the crossing is driven by fast readout fluctuations; near the committed attractors (≈ 3) the slow choice mode governs it, but non-crossers are dropped (HSSM has no censored likelihood). Same pipeline, nine more pooled fits, all converged (R-hat 1.00, one divergence in 36 000 draws).

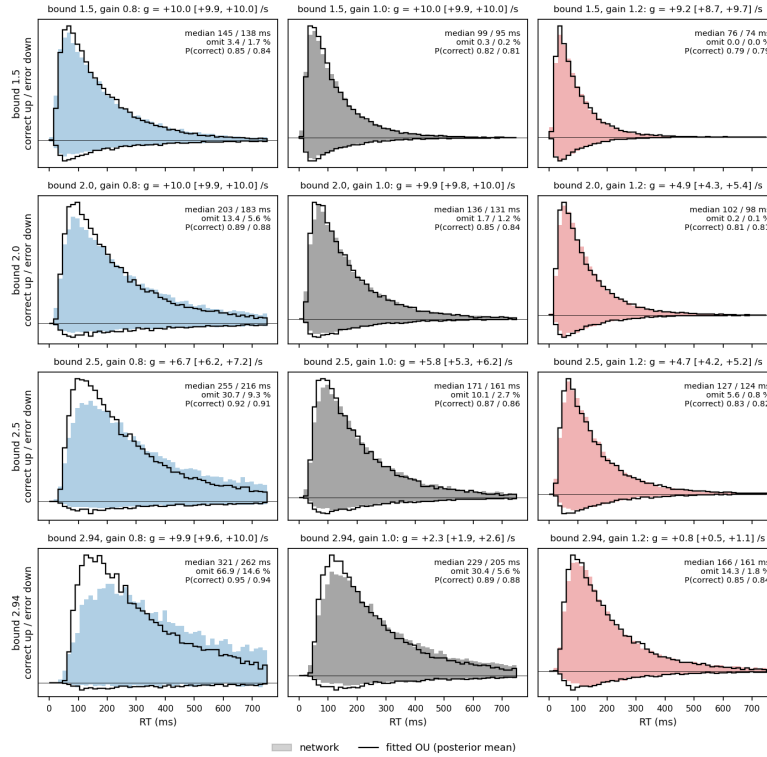


Figure 3. RT posterior predictive at each threshold (rows) and gain (columns), all coherences, correct up and error down, network (filled) vs fitted OU (line). Panel titles: fitted native g with 94 % HDI; insets: median RT, omission rate and accuracy, network / model.

bound	gain	g native (per s) [94 % HDI]	omissions network / model (%)	RT-quantile error (ms)	kernel slope, fitted OU
1.5	0.8	+10.0 [+9.9, +10.0]	3.4 / 1.8	6.5	-0.041
	1.0	+10.0 [+9.9, +10.0]	0.3 / 0.2	1.6	-0.057
	1.2	+9.2 [+8.7, +9.7]	0.0 / 0.0	0.8	-0.064
2.0	0.8	+10.0 [+9.9, +10.0]	13.4 / 6.1	20.9	-0.024
	1.0	+9.9 [+9.8, +10.0]	1.7 / 1.2	3.7	-0.045
	1.2	+4.9 [+4.3, +5.4]	0.2 / 0.2	2.4	-0.057
2.5	0.8	+6.7 [+6.2, +7.2]	30.7 / 11.3	33.7	-0.011
	1.0	+5.8 [+5.3, +6.2]	10.1 / 2.9	10.1	-0.035
	1.2	+4.7 [+4.2, +5.2]	5.6 / 0.6	3.5	-0.052
2.94	0.8	+9.9 [+9.6, +10.0]	66.9 / 20.7	37.8	+0.006
	1.0	+2.3 [+1.9, +2.6]	30.4 / 6.3	24.1	-0.025
	1.2	+0.8 [+0.5, +1.1]	14.3 / 1.8	4.3	-0.043
network	0.8 / 1.0 / 1.2	predicted + / 0 / -			+0.039 / +.002 / -.026

Table 2. Positive g is leaky; every HDI excludes zero; +10.0 is the likelihood's ceiling. Raising the threshold lowers g and orders it by gain as predicted, and the kernel ordering appears too, shifted toward primacy. g never becomes negative, and the fit fails where the ordering appears: the leaky network's RTs have a long flat tail and many omissions that a one-timescale linear OU cannot produce. Gain 1.2 fits at every threshold and reaches near-perfect integration (+0.8) at 2.94, not instability (landscape -6.2).

3. Result: the same OU fitted to each trial's choice given its evidence stream (Brunton-style, no RTs)

$a_{t+1} = a_t + (v e_t - g a_t) dt + \sqrt{dt} \xi_t$, $dt = 1$ ms, choice = sign(a_{750}); e_t is the evidence the network received. Without a bound a_t is Gaussian given the evidence, so each choice has an exact likelihood; fitted by maximum likelihood and NUTS per network and gain (60 fits) and pooled.

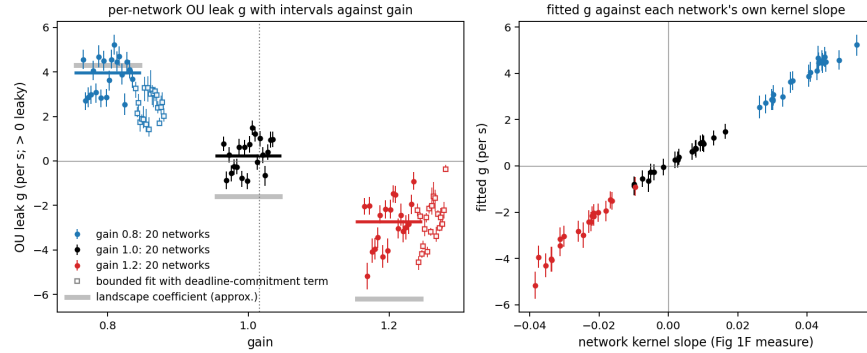


Figure 4. Left: fitted g per network with 94 % intervals (filled), pooled fit (bars), bounded fit with the deadline-commitment term (open squares), landscape drift coefficient (gray), kernel zero crossing of Fig 1F at gain 1.016 (dotted). Right: g against each network's own kernel slope, $r = 0.996$.

gain	median g across networks (per s) [IQR]	intervals excluding 0 on the predicted side	pooled g [94 % HDI]	landscape coefficient (approx.)
0.8	+4.13 [+3.02, +4.63]	20 of 20	+3.95 [+3.83, +4.06]	+4.3
1.0	+0.32 [-0.34, +0.79]	10 positive, 5 negative	+0.23 [+0.15, +0.32]	-1.6
1.2	-2.63 [-3.56, -2.03]	20 of 20	-2.73 [-2.83, -2.62]	-6.2

Table 3. Leaky $\rightarrow$ near zero $\rightarrow$ unstable, in every network at the extreme gains. Zero crossing of g against gain: 1.016 (Fig 1F kernel: 1.016). Recovery on real evidence with fitted or theoretical g planted: 60 of 60 signs, 56 of 60 intervals cover the truth.

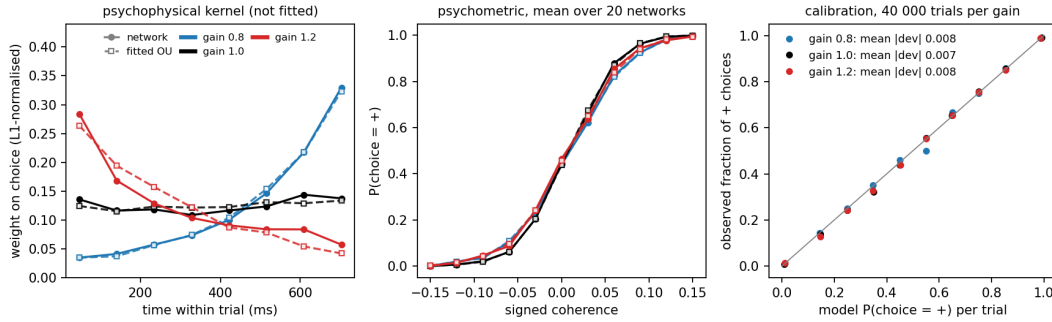


Figure 5. Posterior predictive. Left: the fit reproduces the psychophysical kernel it was not fitted to (slopes +0.041 / +0.002 / -0.031 vs network +0.039 / +0.002 / -0.026). Middle: $P(\text{choice})$ by signed coherence, within 0.015 everywhere. Right: per-trial predicted probability against observed choice fraction.

Caveats. The magnitude at gain 0.8 depends on how the sticky bound is constrained (+4.1 free, +2.5 with the deadline commitment fraction); sign and ordering do not. The linear OU under-reports the repulsion at gain 1.2 (-2.6 vs -6.2) because it cannot saturate into wells. A joint RT-and-choice fit would need a second, fast timescale; the Brody lab's Fokker-Planck grid likelihood is the route to both.